Technical Specification

Power and Cooling Systems | Project DC-PUNE-01 | Revision A

SYNTHETIC DOCUMENT - created for a hackathon demo, not a real project specification.

1. General

1.1 This specification covers the electrical power and mechanical cooling systems of the data centre. Equipment is referred to by its tag, for example UPS-02.

1.2 The facility is designed to Tier III. Clauses 4.14 and 5.11 apply only if the owner raises the target to Tier IV.

4. Electrical power systems

4.1 Main LV switchgear current rating. The main LV switchgear (SWG-01) shall have a rated busbar current of not less than 4000 A.

4.2 UPS rated capacity. Each UPS module (UPS-02) shall have a rated output of not less than 500 kW.

4.3 UPS battery runtime. The UPS battery system (BAT-02) shall support the full rated load for not less than 10 minutes.

4.4 UPS efficiency. UPS-02 shall achieve an efficiency of not less than 96 % in double-conversion mode at full load.

4.5 Generator standby rating. The standby diesel generator (GEN-01) shall have a standby rating of not less than 1500 kW.

4.6 Generator start time. GEN-01 shall start and accept load within not more than 15 seconds of a utility failure signal.

4.7 Generator fuel storage. On-site fuel storage for GEN-01 shall support not less than 48 hours of operation at full load.

4.8 ATS transfer time. The automatic transfer switch (ATS-01) shall complete a transfer within not more than 10 seconds.

4.9 ATS current rating. ATS-01 shall be rated for not less than 2000 A continuous current.

4.10 Feeder breaker rating. The main feeder circuit breaker (BRK-05) shall have a rated current of exactly 630 A.

4.11 Breaker breaking capacity. BRK-05 shall have a short-circuit breaking capacity of not less than 50 kA.

4.12 PDU output voltage. The power distribution unit (PDU-01) shall deliver an output voltage of 415 V.

4.13 Concurrent maintainability. Each critical load shall be fed through not fewer than 2 independent power paths, so that any one path can be isolated for maintenance without load interruption (PDU-01).

4.14 Power fault tolerance. The power distribution (PDU-01) shall tolerate not fewer than 1 simultaneous fault without interrupting any critical load.

5. Mechanical cooling systems

5.1 Chiller cooling capacity. The water-cooled chiller (CHL-01) shall have a cooling capacity of not less than 1200 kW.

5.2 Chiller efficiency. CHL-01 shall achieve a coefficient of performance (COP) of not less than 5.5 at full load.

5.3 Chilled water supply temperature. CHL-01 shall deliver chilled water at a supply temperature of 18 degC.

5.4 Chilled water pump flow. The chilled water pump (PMP-01) shall deliver a flow of not less than 36 L/s.

5.5 Chilled water pump head. PMP-01 shall provide a rated head of not less than 30 m.

5.6 Cooling tower heat rejection. The cooling tower (CT-01) shall have a heat rejection capacity of not less than 1500 kW.

5.7 CRAH cooling capacity. Each CRAH unit (CRAH-01) shall have a cooling capacity of not less than 100 kW.

5.8 CRAH supply air temperature. CRAH-01 shall deliver supply air at a temperature of not more than 25 degC.

5.9 Cooling redundancy (N+1). The cooling plant shall include not fewer than 1 standby chiller, so that full capacity is maintained when any one chiller is out of service (CHL-01).

5.10 Cooling restart after transfer. After a transfer to generator power, CHL-01 shall resume cooling within not more than 120 seconds.

5.11 Cooling fault tolerance. A single failure in the cooling path shall not reduce available cooling capacity below 100 % of the design load (CHL-01).